

7.12 Quiz: Logarithmic and Exponential Functions

Problems 1–2: no calculator. Problems 3–6: calculator allowed.

1. [Maximum mark: 5]

Write each of the following expressions in the form $\ln k$, where $k \in \mathbb{Z}^+$.

(a) $\ln 8 + \ln 9$ [1]

(b) $2 \ln 6 - \ln 4$ [2]

(c) $\ln 48 - 2 \ln 2 - \ln 3$ [2]

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2. [Maximum mark: 5]

(a) Solve the equation $3^{2x} = 27$.

[2]

(b) Solve the equation $\log_2(x+3) + \log_2(x-3) = 4$, where $x > 3$.

[3]

3. [Maximum mark: 6]

Let $g(x) = 2 + \log_3(x - 1)$.

(a) State the equation of the vertical asymptote of the graph of g . [1]

(b) Find the x -intercept of the graph of g . [2]

(c) Find an expression for $g^{-1}(x)$. [3]

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4. [Maximum mark: 6]

An algae bloom is spreading across a small lake. The mass of algae, A , in kilograms, t days after the bloom is first measured, is modelled by

$$A(t) = 8e^{kt}, \quad k > 0.$$

After 4 days, the mass of algae is 32 kg.

- (a) Show that $k = \frac{1}{2} \ln 2$. [2]
- (b) Find the mass of algae after 11 days. Give your answer correct to the nearest integer. [2]
- (c) Find the time, in days, at which the mass of algae first reaches 1000 kg. Give your answer correct to two decimal places. [2]

5. [Maximum mark: 6]

Hugo opens a savings account that pays interest at a nominal annual rate of 4.8%, compounded monthly. He deposits \$5000 and makes no further deposits or withdrawals. Let $V(t)$ be the value of the account, in dollars, t years after the deposit.

- (a) Write down a formula for $V(t)$. [1]
- (b) Find the value of the account at the end of 6 years, correct to the nearest cent. [2]
- (c) Find the smallest number of full years required for the value of the account to first exceed \$8000. [3]

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6. [Maximum mark: 6]

The sound intensity level, L , measured in decibels (dB), of a sound of intensity I , in watts per square metre, is given by

$$L = 10 \log_{10} \left(\frac{I}{I_0} \right),$$

where $I_0 = 10^{-12} \text{ W/m}^2$ is the threshold of human hearing.

- (a) A vacuum cleaner produces a sound of intensity $I = 3.2 \times 10^{-5} \text{ W/m}^2$. Find its sound intensity level, in decibels, correct to one decimal place. [2]
- (b) A jet engine produces a sound intensity level of 130 dB. Find the sound intensity I , in W/m^2 . Give your answer in the form $a \times 10^k$, where $1 \leq a < 10$ and $k \in \mathbb{Z}$. [2]
- (c) Show that doubling the intensity of a sound increases its sound intensity level by approximately 3.01 dB. [2]